



PREGNANCYGUIDEAI

PregnancyGuideAI: An Innovative Solution for Maternal Health

Project Lead:

Fidele Allarate

Email: fidele.allarate@aims-cameroon.org

Tel: +23560856572/+237640044867

Project Lead Assistant:

Denenodji Phoebe

Email : phoebedenenodji@gmail.com

Tel : +23560949472

Project Adviser:

Ngarparis Alfred

Tel : +23560838382

January 28, 2025

Context

In Chad, maternal health is a major challenge, with an extremely high maternal mortality rate (860 deaths per 100,000 births, according to the WHO). The lack of access to medical services, particularly in rural areas, as well as the low number of regular prenatal consultations, are key factors contributing to this dramatic situation. Pregnant women, especially those living in isolated areas, often lack access to qualified midwives or gynecologists, significantly increasing the risks during pregnancy and childbirth. An innovative solution, based on artificial intelligence, could help fill this lack of access to care. By developing an intelligent agent capable of simulating a prenatal consultation and offering tailored medical advice, we could improve access to care, reduce risks and save lives.

Objectives

Develop and deploy an AI-based intelligent agent to provide prenatal consultations, particularly in rural and remote areas of Chad. This system will guide pregnant women, detect the risks of complications and direct them to appropriate care. To do this, we have some specific objectives to achieve:

1. Create an AI-based intelligent agent capable of simulating a prenatal visit and providing personalized medical advice based on the health status of each pregnant woman.
2. Train women and communities on the use of the intelligent agent, so that they can benefit from regular consultations, even without direct access to a health professional.
3. Improve access to specialized care by using AI to detect risks and refer women to health centers or professionals in the event of complications.
4. Raise awareness among communities of the importance of regular prenatal consultations and postnatal care, through appropriate information campaigns.

Planned activities

1. **Development of the intelligent agent (AI):**
 - (a) Creation of a chatbot and a mobile application that uses artificial intelligence to ask questions about pregnancy, analyze the answers and provide tailored advice. The agent will be designed to interact in several local languages, especially the languages spoken in Chad (for the moment).
 - (b) Integration of voice recognition for women who are illiterate or have difficulty writing.
2. **Monitoring and evaluation of maternal health:** The AI agent will monitor symptoms and signs of complications (hypertension, gestational diabetes, infections, etc.) and alert the pregnant woman or a healthcare professional if necessary.
3. **Deployment of the agent in the villages;**
4. **User training:**
 - (a) Organize workshops and training sessions to teach women, community health workers and families how to use the smart agent.
 - (b) Disseminate information through local campaigns via community radio, and awareness meetings.
5. **Integration with local health systems:** Connect the intelligent agent with local health structures, allowing health professionals to monitor the condition of pregnant women remotely and intervene in the event of complications.
6. **Communication campaigns:** Use local channels (radio, SMS, community influencers) to promote the project, explain how it works and encourage women to use the intelligent agent.

Beneficiaries

1. Pregnant women, particularly those living in rural or remote areas of Chad, without access to regular antenatal care.
2. Local communities, who will benefit from a better understanding of pregnancy risks and maternal health practices.
3. Community health workers, who will be trained to use the smart agent to help manage antenatal care in their community.

Budget for a prototype

In three months, we will propose a solution in French, which will be adapted to local languages later. We need:

Title	Number	Hourly rate (FCA)	Day(08 hours)	Monthly rate (FCA)	3 months(CFA)
Chatbot developers	1	10000	80000	2 400 000	7 200 000
Mobile developers	1	10000	80000	2 400 000	7 200 000
Expert gynecologist	1	10000	80000	2 400 000	7 200 000
Deployment	1				4 000 000
Total					25 600 0000

Monetization of the solution

Our economic model is based on several sources of financing and income:

1. Freemium model: Offer access to basic consultation for free, with premium features available (specialized consultations, personalized follow-up) for a subscription.
2. Partnerships with governments and NGOs: Seek grants and funding from international organizations, NGOs, or governments to expand access to this solution in the most vulnerable areas across Africa .
3. Institutional funding model: Local hospitals and clinics can pay for a prenatal care monitoring and management service, using AI to strengthen their remote care capacity.
4. Advertising and sponsorship: Maternal health product companies (vitamins, supplements) can sponsor the application, by displaying non-intrusive advertisements.